

Influence of air pollution on extrinsic childhood asthma.

PMID: 2919801 | Indexed in PubMed

A crossed comparative study was done with 248 extrinsic asthmatic children living either in polluted or non-polluted areas (mean emission per year of sedimentary material greater than or less than 300 mg/m²/day, respectively) to establish the influence of air pollution on childhood extrinsic asthma. The mean number of wheezing crises per year was significantly higher for the children living in polluted areas (10.4 versus 7.69). In addition, incidence of severe asthma (types II, III, and IV) in children living in polluted areas was markedly increased whereas the slight form of asthma (type I) was more frequent in children living in non-polluted areas. No correlation, however, between the wheezing episodes and levels of atmospheric contaminants (fumes and SO₂) was detected when a group of 84 extrinsic asthmatic children living in polluted areas was studied longitudinally for a year. The data indicate that air pollution, as an isolated agent, plays a transient role in the appearance of wheezing episodes in subjects with extrinsic asthma. Results also suggest that the air pollution may potentiate wheezing episodes via alternative mechanisms.

The association between childhood asthma prevalence and monitored air pollutants in metropolitan areas, United States, 2001-2004.

PMID: 20117766 | Indexed in PubMed

Air pollution exposure has been linked to adverse respiratory health outcomes among children, primarily in studies of acute exposures that are often in limited geographic areas. We sought to assess the association between chronic outdoor air pollution exposure, as measured by 12-month averages by county, and asthma among children in metropolitan areas across the nation. Eligible children included those aged 3-17 years residing in US metropolitan areas who were sampled in the 2001-2004 National Health Interview Survey (N=34,073). 12-month average air pollutant levels for sulfur dioxide, nitrogen dioxide, ozone and particulate matter were compiled by county for 2000-2004. Eligible children were linked to pollutant levels for the previous 12 months for their county of residence. Adjusted odds ratios of having current asthma or an asthma attack in the past 12 months were estimated in single pollutant logistic regression models. Children in counties with ozone and, to a less consistent degree, particulate matter levels in the highest quartile were more likely to have current asthma and/or a recent asthma attack than children residing in counties with the lowest pollution levels; the adjusted odds for current asthma for the highest quartile of estimated ozone exposure was 1.56 (95% confidence interval [CI]: 1.15, 2.10) and for recent asthma attack 1.38 (95% CI: 0.99, 1.91). No associations were found with sulfur dioxide or nitrogen dioxide levels. Although the current US standard for ozone is based on short-term exposure, this cross-sectional study suggests that chronic (12-month) exposure to ozone and particles is related to asthma outcomes among children in metropolitan areas throughout the US.

Role of urbanization and air pollution in adolescent asthma: a mass screening in Taiwan.

PMID: 11760369 | Indexed in PubMed

The prevalence of asthma in school children in Taiwan is increasing. This study used mass screening

among middle school children in Taiwan to determine the prevalence of asthma and related factors. Data were collected from parents using a self-reported questionnaire and from children using the International Study of Asthma and Allergies in Childhood (ISAAC) video questionnaire. Six study teams conducted the survey nationwide in 1995-1996, with the assistance of middle school nurses and teachers. Among the 1,018,031 students at 795 middle schools who returned questionnaires, 8.5% had a history of asthma (ranging in prevalence from 4.2% to 13% in 25 areas). The prevalence of asthma was higher in boys than in girls (10.0% vs 7%) and was highest in more urbanized areas (11.2%), followed by moderately urbanized areas (7.4%) and less urbanized and rural areas (6.5%). Controlling for age, family smoking, family incense burning, and parental education level, multivariate logistic regression models indicated that children living in an area with heavy air pollution were more likely to have asthma than those in an area with no or light pollution (odds ratio, OR = 2.01 and 95% confidence interval, CI = 1.94-2.09 based on parental ranking of pollution level, or OR = 1.30 and 95% CI = 1.18-1.42 based on pollution level reported by the Environmental Protection Administration). Adolescent asthma in Taiwan is most prevalent in the most urbanized areas and decreases in prevalence in less urbanized areas. This study also found that higher parental education level and higher area air pollution were associated with higher adolescent asthma prevalence.

Assessing the impact of air pollution on childhood asthma morbidity: how, when, and what to do.

PMID: 29493555 | Indexed in PubMed

Exposure to air pollutants is linked with poor asthma control in children and represents a potentially modifiable risk factor for impaired lung function, rescue medication use, and increased asthma-related healthcare utilization. Identification of the most relevant pollutants to asthma as well as susceptibility factors and strategies to reduce exposure are needed to improve child health. The current available literature supports the association between pollutants and negative asthma outcomes. Ethnicity, socioeconomic status, and presence of certain gene polymorphisms may impact susceptibility to the negative health effects of air pollution. Improved air quality standards were associated with better asthma outcomes. The link between air pollution and pediatric asthma morbidity is supported by the recent relevant literature. Continued efforts are needed to identify the most vulnerable populations and develop strategies to reduce exposures and improve air quality.

Hystero-Salpingography in current clinical practice-old flames, die hard!

PMID: 38913135 | Indexed in PubMed

Hysterosalpingography (HSG) remains a valuable diagnostic tool in current clinical practice, offering crucial insights into endometrial cavity, fallopian tubes and the adjoining part of the pelvic peritoneal cavity. Despite the emergence of alternative imaging and non-imaging options, HSG continues to be widely utilized due to its diagnostic accuracy, cost-effectiveness, and easy accessibility. Due attention to the correct technique and optimal image interpretation will further enhance its diagnostic accuracy and precision in the work-up of patients with fertility problems.

HYSTEROSALPINGOGRAPHY AND LAPAROSCOPY IN EVALUATING FALLOPIAN TUBES IN THE MANAGEMENT OF INFERTILITY IN COTONOU, BENIN REPUBLIC.

PMID: 26587523 | Indexed in PubMed

Hysterosalpingography, along with laparoscopy, are the most requested examinations for tubal factor exploration for infertility, in developing countries. To compare the results of hysterosalpingography and laparoscopy in patients assessed for infertility. This was a 5 years retrospective, descriptive study done at the Obstetrics and Gynecology Clinic of the HKM Centre, National University Hospital. All the patients admitted for infertility of tubal origin were included. These included 96 patients who had undergone hysterosalpingography followed by laparoscopy. The analysis was done with the SPSS version 12.0.1. The mean age of the patients was 33.3 years. Infertility was primary in 66.3% of cases and secondary in 33.7% of cases and the average duration was 48.9 months. Hysterosalpingography diagnosed 9.37% of proximal tubal obstruction while laparoscopy diagnosed same in 17.71%. Besides pelvic adhesive bands seen in 33.33% of cases, laparoscopy was able to visualize patent tubes with some pathology in 11.46%, and pelvic endometriosis in 6.25% of cases. The results of HSG and those of laparoscopy are complementary in tubal infertility evaluation. While HSG seems to be reliable when the tubes are patent, laparoscopy helps to reveal false tubal obstructions observed with HSG, and also helps in the diagnosis of pelvic adhesive bands and endometriosis.

Comparison of four different pain relief methods during hysterosalpingography: a randomized controlled study.

PMID: 25848848 | Indexed in PubMed

Hysterosalpingography (HSG) is the most commonly used method for evaluating the anatomy and patency of the uterine cavity and fallopian tubes, and is an important tool in the evaluation of infertility. The most frequent side effect is the pain associated with the procedure. To evaluate four analgesic methods to determine the most useful method for reducing discomfort associated with HSG. In the present prospective study, 75 patients undergoing HSG for evaluation of infertility were randomly assigned to four groups: 550 mg of a nonsteroidal anti-inflammatory drug (NSAID) (group 1); 550 mg NSAID + paracervical block (group 2); 550 mg NSAID + paracervical analgesic cream (group 3); or 550 mg NSAID + intrauterine analgesic instillation (group 4). A visual analogue scale was used to assess the pain perception at five predefined steps. Instillation of the liquids used for HSG was found to be the most painful step of HSG, and this step was where the only significant difference among groups was observed. When comparing visual analogue scale scores, group 2 and group 3 reported significantly less pain than the other groups. Group 1 reported significantly higher mean ($\pm$ SD) scores (7.2 ± 1.6) compared with groups 2 and 3 (4.7 ± 2.5 and 3.8 ± 2.4 , respectively) ($P < 0.001$). In addition, group 2 reported significantly less pain than group 4 (4.7 ± 2.5 versus 6.7 ± 1.8 , respectively) ($P < 0.02$). For effective pain relief during HSG, in addition to 550 mg NSAID, local application of lidocaine cream to the posterior fornix of the cervix uteri and paracervical lidocaine injection into the cervix uteri appear to be the most effective methods.

Hysterosalpingography: a reemerging study.

PMID: 16549607 | Indexed in PubMed

Hysterosalpingography (HSG) has become a commonly performed examination due to recent advances and improvements in, as well as the increasing popularity of, reproductive medicine. HSG plays an important role in the evaluation of abnormalities related to the uterus and fallopian tubes. Uterine abnormalities that can be detected at HSG include congenital anomalies, polyps, leiomyomas, surgical changes, synechiae, and adenomyosis. Tubal abnormalities that can be detected include tubal occlusion, salpingitis isthmica nodosum, polyps, hydrosalpinx, and peritubal adhesions. Some complications can occur with HSG-most notably, bleeding and infection-and awareness of the possible complications of HSG is essential. Nevertheless, HSG remains a valuable tool in the evaluation of the uterus and fallopian tubes. Radiologists should become familiar with HSG technique and the interpretation of HSG images.

Hysterosalpingography: technique and applications.

PMID: 19632497 | Indexed in PubMed

Hysterosalpingography (HSG) remains an important radiologic procedure in the investigation of infertility and has become a commonly performed examination due to recent advances of reproductive medicine. HSG demonstrates the morphology of the uterine cavity, the lumina, and the patency of the fallopian tubes. In this review article, we present the technical parameters of the examination, indications, contraindications, and possible complications of HSG. We also illustrate a variety of abnormalities of the uterus and fallopian tubes that can be detected accurately with HSG. We believe that, with the increased demand for HSG, radiologists should be familiar with HSG technique and the interpretation of HSG images.

The categorization of opaque pathologies outside of contrast media in hysterosalpingography which facilitate interpretation: A pictorial review.

PMID: 38868446 | Indexed in PubMed

Hysterosalpingography (HSG) is a practical and reliable imaging method to evaluate the cervical canal, uterine isthmus, uterine cavity, and fallopian tubes. Using HSG, opaque pathologies outside of contrast media can be detected as well as pathologies of uterus and fallopian tubes. We aim to present and categorize some uncommon and interesting abnormal findings that are located outside of the contrast areas in HSG. This is a pictorial review that depicts various types of HSG images that include opaque pathologies outside of the contrast areas. Images have been extracted from valuable archives collected over 50 yr by professor Shahrzad. A plain pelvic film contains soft tissues of the pelvis, bony structures, artifacts, or foreign bodies. Categorization might easily help the radiologist to interpret the HSG cliché. Opaque pathologies outside of contrast area in HSG can be categorized into 2 groups: "Pelvic Tissue Related" and "Foreign Bodies". Pelvic tissue abnormalities might have a gynecologic or non-gynecologic source. Foreign bodies can be located in the pelvis or outside of the body. HSG is a reliable and inexpensive procedure. Familiarity with the pathologies of pelvic tissues and the accurate interpretation of HSG images are important.